

Python DA Assignment 2 - Data Visualization

1. Load the dataset

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Dataset: Seaborn taxi's (NYC taxi trips, March 2019) Tools: Pandas, Matplotlib, Seaborn

1. Load the dataset

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

# Load the 'taxi' dataset
df = sns.load_dataset("taxi")

print("Shape:", df.shape)
df.head()
```

Shape: (6433, 14)

	pickup	dropoff	passengers	distance	fare	tip	tolls	total	color	payment	pickup_zone	dropoff_zone	pickup_borough	dropoff_borough
0	2019-03-20 21:09	2019-03-20 21:24	1	1.60	7.0	2.15	0.0	12.95	yellow	credit card	Lenox Hill West	UN/Turtle Bay South	Manhattan	Manhattan
1	2019-03-16 11:55	2019-03-16 11:50	1	0.79	5.0	0.00	0.0	9.30	yellow	cash	Upper West Side South	Upper West Side South	Manhattan	Manhattan
2	2019-03-17 53:01	2019-03-18 00:25	1	1.37	7.5	2.36	0.0	14.16	yellow	credit card	Alphabet City	West Village	Manhattan	Manhattan
3	2019-03-10 01:23:59	2019-03-10 01:49:51	1	7.70	27.0	6.15	0.0	36.95	yellow	credit card	Hudson Sq	Yorkville West	Manhattan	Manhattan
4	2019-03-13 27:42	2019-03-13 37:14	3	2.16	9.0	1.10	0.0	13.40	yellow	credit card	Midtown East	Yorkville West	Manhattan	Manhattan

Next steps: [Generate code with df](#) [New interactive sheet](#)

Variables

Name	Type	Shape	Value
df	DataFrame	(6433, 14)	pickup, dropoff, passengers, distance, fare, tip, tolls, total, color, payment, pickup_zone, dropoff_zone, pickup_borough, dropoff_borough

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2	2019-03-17 53:01	2019-03-18 00:25	1	1.37	7.5	2.36	0.0	14.16	yellow	credit card	Alphabet City	West Village	Manhattan	Manhattan
3	2019-03-10 01:23:59	2019-03-10 01:49:51	1	7.70	27.0	6.15	0.0	36.95	yellow	credit card	Hudson Sq	Yorkville West	Manhattan	Manhattan
4	2019-03-13 27:42	2019-03-13 37:14	3	2.16	9.0	1.10	0.0	13.40	yellow	credit card	Midtown East	Yorkville West	Manhattan	Manhattan

Next steps: [Generate code with df](#) [New interactive sheet](#)

Variables

How should I handle the missing values in the dataset?

The notebook already contains a detailed explanation of how missing values are handled. In summary, for 'pickup_zone' and 'pickup_borough', rows with missing values were dropped because these columns are central to the analysis and cannot be reasonably guessed. For 'payment', 'dropoff_zone', and 'dropoff_borough', missing values were imputed using the mode, as these are categorical columns with a small percentage of missing data. You can see the full explanation in the 'Handle missing values' section of the notebook.

[Create a bar chart showing the count of taxis by pickup_borough](#)

[Calculate the correlation between distance and fare using the clean dataset](#)

[Generate a box plot of fares for different payment methods](#)

What can I help you build?

Gemini 2.5 Flash 8%

2. Handle missing values

2. Handle missing values

```

# Missing values per column
missing = df.isnull().sum()
missing = missing[missing > 0]
print("Columns with missing data:\n")
print(pd.DataFrame({"missing": missing, "pct": (missing / len(df) * 100)})

```

Columns with missing data:

	missing	pct
payment	44	0.68
pickup_zone	26	0.40
dropoff_zone	45	0.70
pickup_borough	26	0.40
dropoff_borough	45	0.70

All missing values are in **categorical** columns. The strategy per column:

Column	Missing	Strategy
payment	44	Mode imputation
dropoff_zone, dropoff_borough	45	Mode imputation
pickup_zone, pickup_borough	26	Drop rows

Reasons: Categorical, small share; mode imputation; Not used as an analysis axis; Central to the borough/zone

```

df_clean = df.copy()

# Critical columns: drop rows with missing pickup location
df_clean = df_clean.dropna(subset=["pickup_zone", "pickup_borough"])

# Remaining categorical columns: impute with the mode
for col in ["payment", "dropoff_zone", "dropoff_borough"]:
    df_clean[col] = df_clean[col].fillna(df_clean[col].mode()[0])

df_clean = df_clean.reset_index(drop=True)

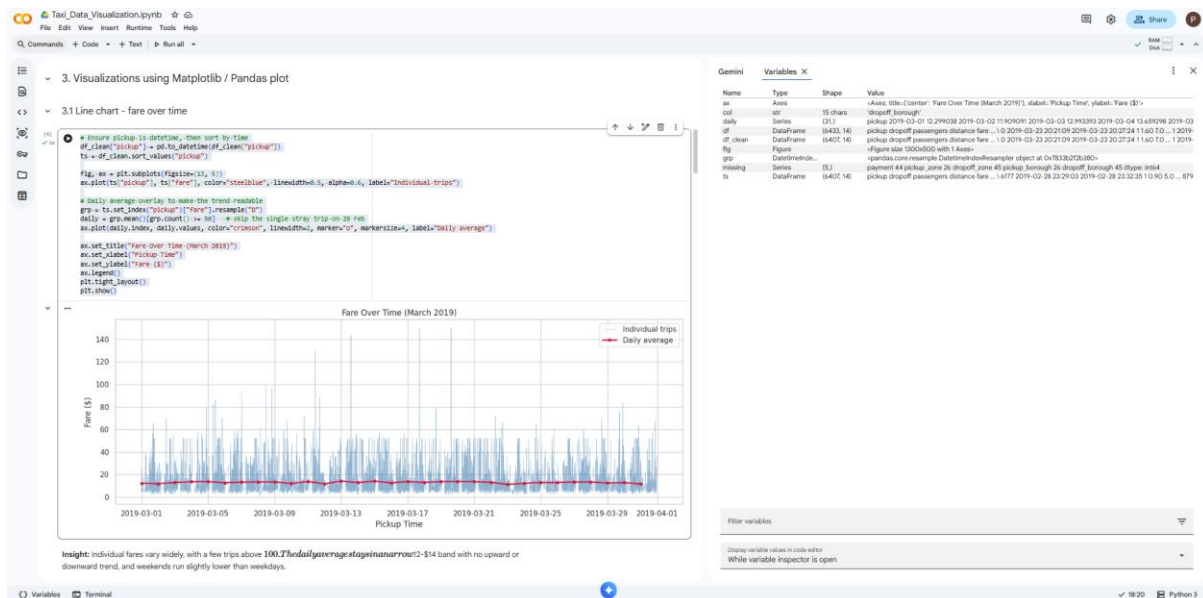
print("Rows before:", len(df), "| Rows after:", len(df_clean))
print("Remaining missing values:", df_clean.isnull().sum().sum())

```

Rows before: 6433 | Rows after: 6407
Remaining missing values: 0

3. Visualizations using Matplotlib / Pandas plot

3.1 Line chart - fare over time

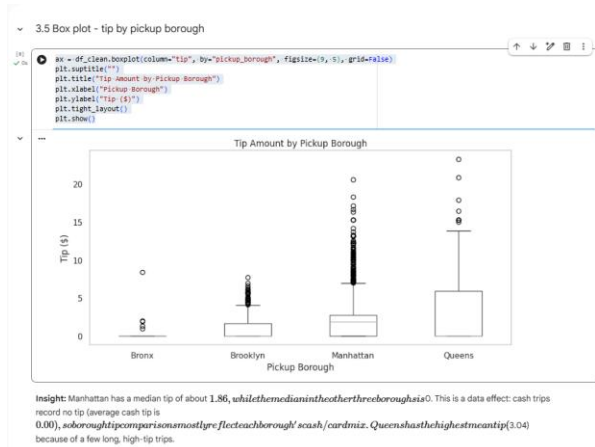


3.2 Bar chart - total fare by pickup borough

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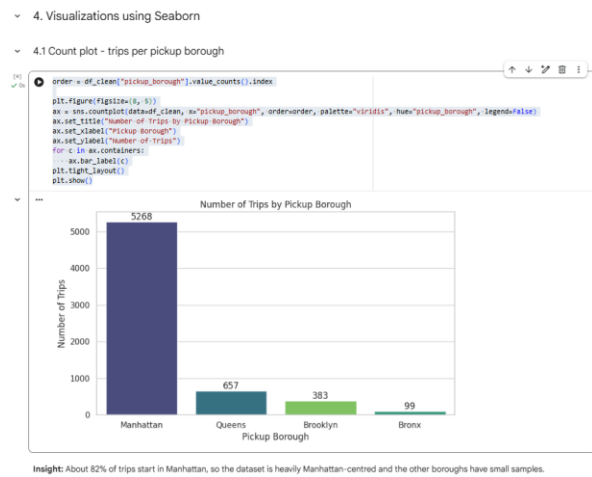
Name	Type	Shape	Value
as	Axis		<Axes title='center': 'Total Fare by Pickup Borough', xlabel='Pickup Borough', ylabel='Total Fare (\$)'>
borough_fare	Series	(4)	pickup_borough Manhattan 58753.42 Queens 16382.06 Brooklyn 6327.48 Bronx 2078.91 Name: fare, dtype: float64
col	str	15 chars	'dropoff_borough'
daily	Series	(21)	pickup 2019-03-01 12:299038 2019-03-02 11909091 2019-03-03 12:993393 2019-03-04 13:659298 2019-03-05 14:327241 2019-03-06 15:116070 ... 1 2019-03-22 20:2724 2019-03-23 20:2724 2019-03-24 20:2724 2019-03-25 20:2724 2019-03-26 20:2724 2019-03-27 20:2724 2019-03-28 20:2724 2019-03-29 20:2724 2019-03-30 20:2724 2019-03-31 20:2724 2019-04-01 20:2724 2019-04-02 20:2724 2019-04-03 20:2724 2019-04-04 20:2724 2019-04-05 20:2724 2019-04-06 20:2724 2019-04-07 20:2724 2019-04-08 20:2724 2019-04-09 20:2724 2019-04-10 20:2724 2019-04-11 20:2724 2019-04-12 20:2724 2019-04-13 20:2724 2019-04-14 20:2724 2019-04-15 20:2724 2019-04-16 20:2724 2019-04-17 20:2724 2019-04-18 20:2724 2019-04-19 20:2724 2019-04-20 20:2724 2019-04-21 20:2724 2019-04-22 20:2724 2019-04-23 20:2724 2019-04-24 20:2724 2019-04-25 20:2724 2019-04-26 20:2724 2019-04-27 20:2724 2019-04-28 20:2724 2019-04-29 20:2724 2019-04-30 20:2724 2019-05-01 20:2724 2019-05-02 20:2724 2019-05-03 20:2724 2019-05-04 20:2724 2019-05-05 20:2724 2019-05-06 20:2724 2019-05-07 20:2724 2019-05-08 20:2724 2019-05-09 20:2724 2019-05-10 20:2724 2019-05-11 20:2724 2019-05-12 20:2724 2019-05-13 20:2724 2019-05-14 20:2724 2019-05-15 20:2724 2019-05-16 20:2724 2019-05-17 20:2724 2019-05-18 20:2724 2019-05-19 20:2724 2019-05-20 20:2724 2019-05-21 20:2724 2019-05-22 20:2724 2019-05-23 20:2724 2019-05-24 20:2724 2019-05-25 20:2724 2019-05-26 20:2724 2019-05-27 20:2724 2019-05-28 20:2724 2019-05-29 20:2724 2019-05-30 20:2724 2019-05-31 20:2724 2019-06-01 20:2724 2019-06-02 20:2724 2019-06-03 20:2724 2019-06-04 20:2724 2019-06-05 20:2724 2019-06-06 20:2724 2019-06-07 20:2724 2019-06-08 20:2724 2019-06-09 20:2724 2019-06-10 20:2724 2019-06-11 20:2724 2019-06-12 20:2724 2019-06-13 20:2724 2019-06-14 20:2724 2019-06-15 20:2724 2019-06-16 20:2724 2019-06-17 20:2724 2019-06-18 20:2724 2019-06-19 20:2724 2019-06-20 20:2724 2019-06-21 20:2724 2019-06-22 20:2724 2019-06-23 20:2724 2019-06-24 20:2724 2019-06-25 20:2724 2019-06-26 20:2724 2019-06-27 20:2724 2019-06-28 20:2724 2019-06-29 20:2724 2019-06-30 20:2724 2019-07-01 20:2724 2019-07-02 20:2724 2019-07-03 20:2724 2019-07-04 20:2724 2019-07-05 20:2724 2019-07-06 20:2724 2019-07-07 20:2724 2019-07-08 20:2724 2019-07-09 20:2724 2019-07-10 20:2724 2019-07-11 20:2724 2019-07-12 20:2724 2019-07-13 20:2724 2019-07-14 20:2724 2019-07-15 20:2724 2019-07-16 20:2724 2019-07-17 20:2724 2019-07-18 20:2724 2019-07-19 20:2724 2019-07-20 20:2724 2019-07-21 20:2724 2019-07-22 20:2724 2019-07-23 20:2724 2019-07-24 20:2724 2019-07-25 20:2724 2019-07-26 20:2724 2019-07-27 20:2724 2019-07-28 20:2724 2019-07-29 20:2724 2019-07-30 20:2724 2019-07-31 20:2724 2019-08-01 20:2724 2019-08-02 20:2724 2019-08-03 20:2724 2019-08-04 20:2724 2019-08-05 20:2724 2019-08-06 20:2724 2019-08-07 20:2724 2019-08-08 20:2724 2019-08-09 20:2724 2019-08-10 20:2724 2019-08-11 20:2724 2019-08-12 20:2724 2019-08-13 20:2724 2019-08-14 20:2724 2019-08-15 20:2724 2019-08-16 20:2724 2019-08-17 20:2724 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20:2724 2019-10-10 20:2724 2019-10-11 20:2724 2019-10-12 20:2724 2019-10-13 20:2724 2019-10-14 20:2724 2019-10-15 20:2724 2019-10-16 20:2724 2019-10-17 20:2724 2019-10-18 20:2724 2019-10-19 20:2724 2019-10-20 20:2724 2019-10-21 20:2724 2019-10-22 20:2724 2019-10-23 20:2724 2019-10-24 20:2724 2019-10-25 20:2724 2019-10-26 20:2724 2019-10-27 20:2724 2019-10-28 20:2724 2019-10-29 20:2724 2019-10-30 20:2724 2019-10-31 20:2724 2019-11-01 20:2724 2019-11-02 20:2724 2019-11-03 20:2724 2019-11-04 20:2724 2019-11-05 20:2724 2019-11-06 20:2724 2019-11-07 20:2724 2019-11-08 20:2724 2019-11-09 20:2724 2019-11-10 20:2724 2019-11-11 20:2724 2019-11-12 20:2724 2019-11-13 20:2724 2019-11-14 20:2724 2019-11-15 20:2724 2019-11-16 20:2724 2019-11-17 20:2724 2019-11-18 20:2724 2019-11-19 20:2724 2019-11-20 20:2724 2019-11-21 20:2724 2019-11-22 20:2724 2019-11-23 20:2724 2019-11-24 20:2724 2019-11-25 20:2724 2019-11-26 20:2724 2019-11-27 20:2724 2019-11-28 20:2724 2019-11-29 20:2724 2019-11-30 20:2724 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Name	Type	Shape	Value
ax	Axes	(4,)	<Axes: title='center: "Tip Amount by Pickup Borough"', label='Pickup Borough', ylabel='Tip (\$)'>
borough_fare	Series	(4,)	pickup_borough Manhattan 58753.42 Queens 16382.06 Brooklyn 6327.48 Bronx 2078.91 Name: fare, dtype: float64
col	str	15 chars	'dropoff_borough'
c	BarContainer	1 item	<BarContainer object of 1 artists>
daily	Series	(31,)	pickup 2019-03-01 12.299038 2019-03-02 11.909091 2019-03-03 12.993393 2019-03-04 13.692978 2019-03-05 13.763028
df	DataFrame	(6433, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
df_clean	DataFrame	(6402, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
fig	Figure	<Figure size 100x500 with 1 Axes>	<Figure size 100x500 with 1 Axes>
fig	DatetimeIndex	3	<pandas.core.resample.DatetimeIndexResampler object at 0x783b2f2b380>
i	int		
missing	Series	(5,)	payment 44 pickup zone 26 dropoff zone 45 pickup_borough 26 dropoff_borough 45 dtype: int64
order	Index	(4,)	Index(['Manhattan', 'Queens', 'Brooklyn', 'Bronx'], dtype='object', name='pickup_borough')
payment_cou...	Series	(2,)	payment credit card 4600 cash 1807 Name: count, dtype: int64
ts	DataFrame	(6402, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
v	float64		<pandas.core.resample.DatetimeIndexResampler object at 0x783b2f2b380>

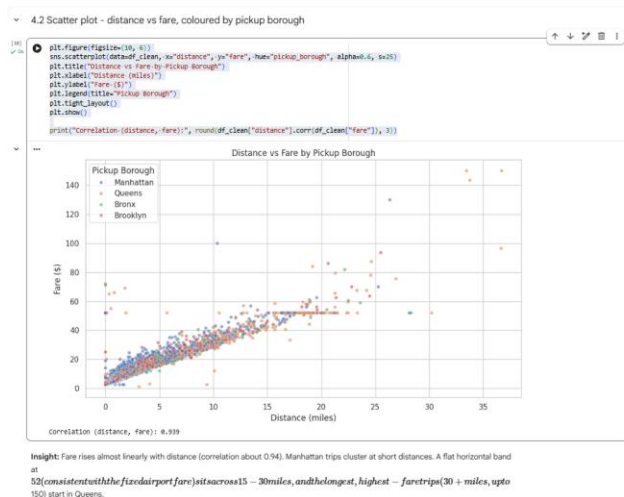
4. Visualizations using Seaborn

4.1 Count plot - trips per pickup borough



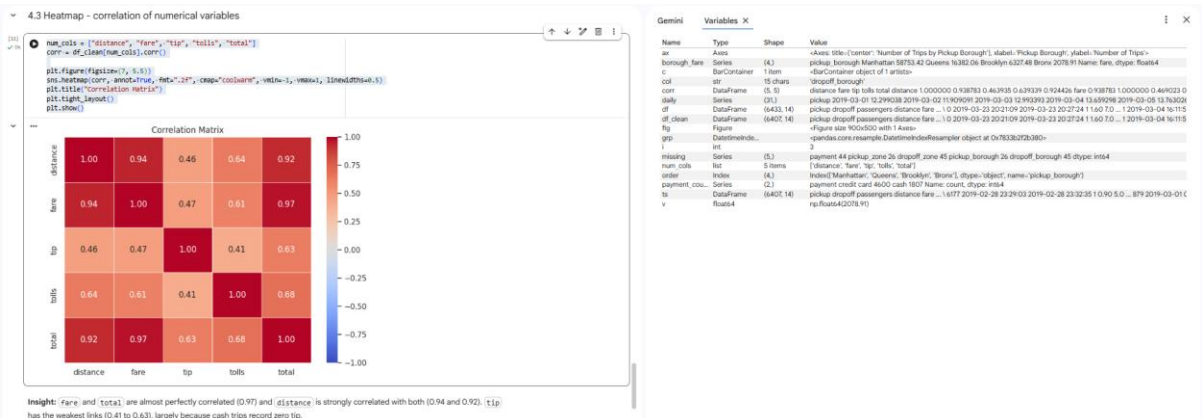
Name	Type	Shape	Value
ax	Axes	(4,)	<Axes: title='center: "Number of Trips by Pickup Borough"', label='Pickup Borough', ylabel='Number of Trips'>
borough_fare	Series	(4,)	pickup_borough Manhattan 58753.42 Queens 16382.06 Brooklyn 6327.48 Bronx 2078.91 Name: fare, dtype: float64
c	BarContainer	1 item	<BarContainer object of 1 artists>
col	str	15 chars	'dropoff_borough'
daily	Series	(31,)	pickup 2019-03-01 12.299038 2019-03-02 11.909091 2019-03-03 12.993393 2019-03-04 13.692978 2019-03-05 13.763028
df	DataFrame	(6433, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
df_clean	DataFrame	(6402, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
fig	Figure	<Figure size 100x500 with 1 Axes>	<Figure size 100x500 with 1 Axes>
fig	DatetimeIndex	3	<pandas.core.resample.DatetimeIndexResampler object at 0x783b2f2b380>
i	int		
missing	Series	(5,)	payment 44 pickup zone 26 dropoff zone 45 pickup_borough 26 dropoff_borough 45 dtype: int64
order	Index	(4,)	Index(['Manhattan', 'Queens', 'Brooklyn', 'Bronx'], dtype='object', name='pickup_borough')
payment_cou...	Series	(2,)	payment credit card 4600 cash 1807 Name: count, dtype: int64
ts	DataFrame	(6402, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
v	float64		<pandas.core.resample.DatetimeIndexResampler object at 0x783b2f2b380>

4.2 Scatter plot - distance vs fare, coloured by pickup borough

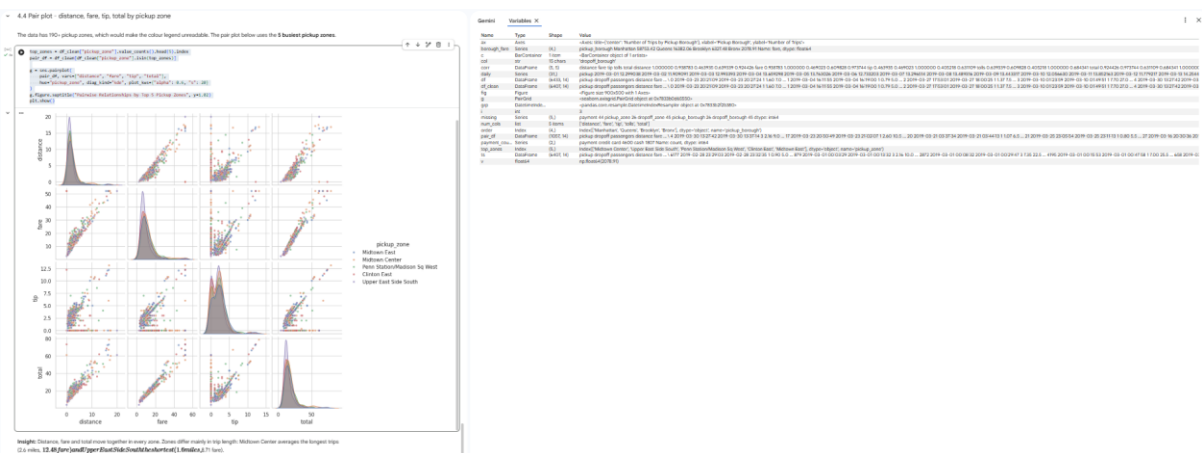


Name	Type	Shape	Value
ax	Axes	(4,)	<Axes: title='center: "Number of Trips by Pickup Borough"', label='Pickup Borough', ylabel='Number of Trips'>
borough_fare	Series	(4,)	pickup_borough Manhattan 58753.42 Queens 16382.06 Brooklyn 6327.48 Bronx 2078.91 Name: fare, dtype: float64
c	BarContainer	1 item	<BarContainer object of 1 artists>
col	str	15 chars	'dropoff_borough'
daily	Series	(31,)	pickup 2019-03-01 12.299038 2019-03-02 11.909091 2019-03-03 12.993393 2019-03-04 13.692978 2019-03-05 13.763028
df	DataFrame	(6433, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
df_clean	DataFrame	(6402, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
fig	Figure	<Figure size 100x500 with 1 Axes>	<Figure size 100x500 with 1 Axes>
fig	DatetimeIndex	3	<pandas.core.resample.DatetimeIndexResampler object at 0x783b2f2b380>
i	int		
missing	Series	(5,)	payment 44 pickup zone 26 dropoff zone 45 pickup_borough 26 dropoff_borough 45 dtype: int64
order	Index	(4,)	Index(['Manhattan', 'Queens', 'Brooklyn', 'Bronx'], dtype='object', name='pickup_borough')
payment_cou...	Series	(2,)	payment credit card 4600 cash 1807 Name: count, dtype: int64
ts	DataFrame	(6402, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:24 1.160 7.0 ... 1 2019-03-04 16:11:5
v	float64		<pandas.core.resample.DatetimeIndexResampler object at 0x783b2f2b380>

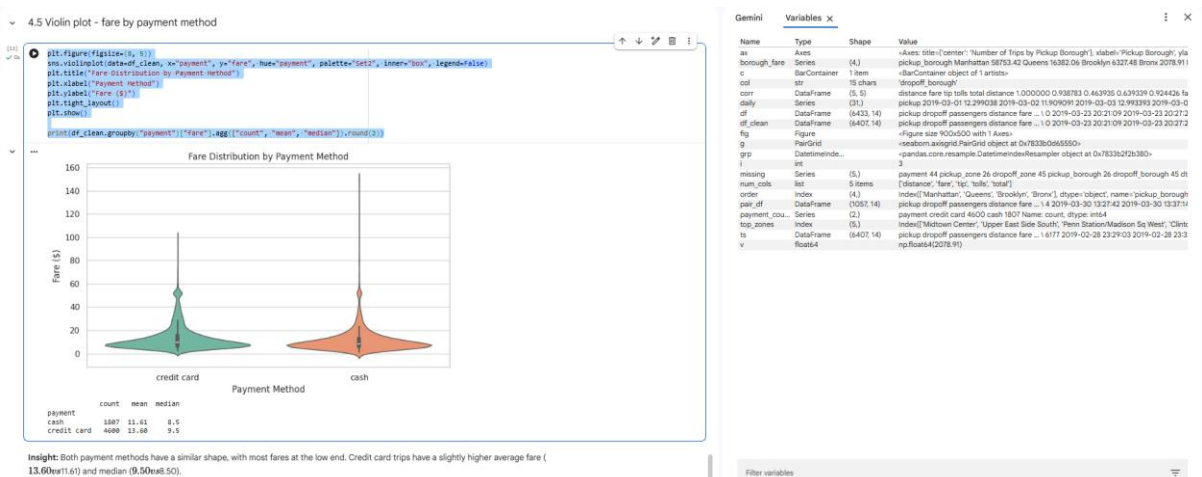
4.3 Heatmap - correlation of numerical variables



4.4 Pair plot - distance, fare, tip, total by pickup zone



4.5 Violin plot - fare by payment method



5. Key findings summary

Taxi_Data_Visualization.py
File Edit View Insert Runtime Tools Help
Commands Code Text Run All

5. Key findings summary

```
summary = {
    "Rows analysed": len(df_clean),
    "Share of trips starting in Manhattan": f"({df_clean['pickup_borough'] == 'Manhattan'}.mean()).1%",
    "Share paid by credit card": f"({df_clean['payment'] == 'credit card'}.mean()).1%",
    "Median distance (miles)": round(df_clean['distance'].median(), 2),
    "Median fare ($)": round(df_clean['fare'].median(), 2),
    "Average tip ($)": round(df_clean['tip'].mean(), 2),
    "Distance-fare correlation": round(df_clean['distance'].corr(df_clean['fare']), 3),
}
pd.Series(summary, name='value').to_frame()
```

	value
Rows analysed	6407
Share of trips starting in Manhattan	82.2%
Share paid by credit card	71.8%
Median distance (miles)	1.65
Median fare (\$)	9.5
Average tip (\$)	1.97
Distance-fare correlation	0.939

Gemini Variables X

Name	Type	Shape	Value
ax	Axes		<Axes: title=[center: Number of Trips by Pickup Borough], xlabel=Pickup Borough, yla
borough_fare	Series	(4,)	pickup_borough Manhattan 58753.42 Queens 16382.06 Brooklyn 6327.48 Bronx 2078.91
c	BarContainer	1 item	<BarContainer object of 1 artists>
col	str	15 chars	'dropoff_borough'
corr	DataFrame	(5, 5)	distance fare tip tolls total distance 1.000000 0.938783 0.463995 0.439239 0.104436 fa
daily	Series	(21,)	pickup 2019-03-01 12:29:03 2019-03-02 11:50:59 2019-03-03 12:39:33 2019-03-0
df	DataFrame	(6433, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:2
df_clean	DataFrame	(6407, 14)	pickup dropoff passengers distance fare ... 1.0 2019-03-23 20:21:09 2019-03-23 20:27:2
fig	Figure		<Figure size 900x500 with 1 Axes>
g	PairGrid		<seaborn.axisgrid.PairGrid object at 0x7833b0d65550>
grp	DatetimeIndex		<pandas.core.resample.DatetimeIndexResampler object at 0x7833b272b380>
i	int	3	
missing	Series	(5,)	payment 44 pickup_zone 26 dropoff_zone 45 pickup_borough 26 dropoff_borough 45 dt
num_cols	list	5 items	['distance', 'fare', 'tip', 'tolls', 'total']
order	Index	(4,)	Index(['Manhattan', 'Queens', 'Brooklyn', 'Bronx'], dtype='object', name='pickup_borough
pair_df	DataFrame	(1057, 14)	pickup dropoff passengers distance fare ... 1.4 2019-03-30 13:27:42 2019-03-30 13:31:14
payment_cou...	Series	(2,)	payment credit card 4600 cash 1807 Name: count, dtype: int64
summary	dict		['Rows analysed': 6407, 'Share of trips starting in Manhattan': 82.2%, 'Share paid by cre
top_zones	Index	(5,)	Index(['Midtown Center', 'Upper East Side South', 'Penn Station/Madison Sq West', 'Cint
ts	DataFrame	(6407, 14)	pickup dropoff passengers distance fare ... 1.6177 2019-02-28 23:29:03 2019-02-28 23:3
y	float64		np.float64(2078.91)

Filter variables

Display variable values in code editor
While variable inspector is open

Variables

Terminal

18:28 Python 3